



PHYS121 Integrated Science-Physics

W5T4 Temperature and Heat

References:

- [1] David Halliday, Jearl Walker, Resnick Jearl, 'Fundamentals of Physics', (Wiley, 2018)
- [2] Doug Giancoli, 'Physics for Scientists and Engineers with modern physics', (Pearson, 2009)
- [3] Hugh D. Young, Roger A. Freedman, 'University Physics with Modern Physics', (Pearson, 2012)

And others specified when needed.



14.4.5. A water well is dug to a depth of 15 meters. A pump is used to bring the water upward through a pipe, but the pump can only raise the water about 7.5 meters. By which of the following methods can the water be brought all the way to the surface?

- a) Connect a second, identical pump in series directly with the first pump.
- b) Connect a second, identical pump in parallel directly with the first pump.
- c) Replace the pump with a more powerful pump.
- d) Replace the pipe with a narrower pipe.
- e) None of these methods will work.

14.7.7. A boy is playing in the bathtub with a toy boat floating on the surface of the water. His favorite rock is the cargo on the boat. If the boy takes the rock from the boat and drops it into the water, how will the water level of the bathtub change, if at all?

- a) The water level will rise.
- b) The water level will fall.
- c) The water level will remain the same.

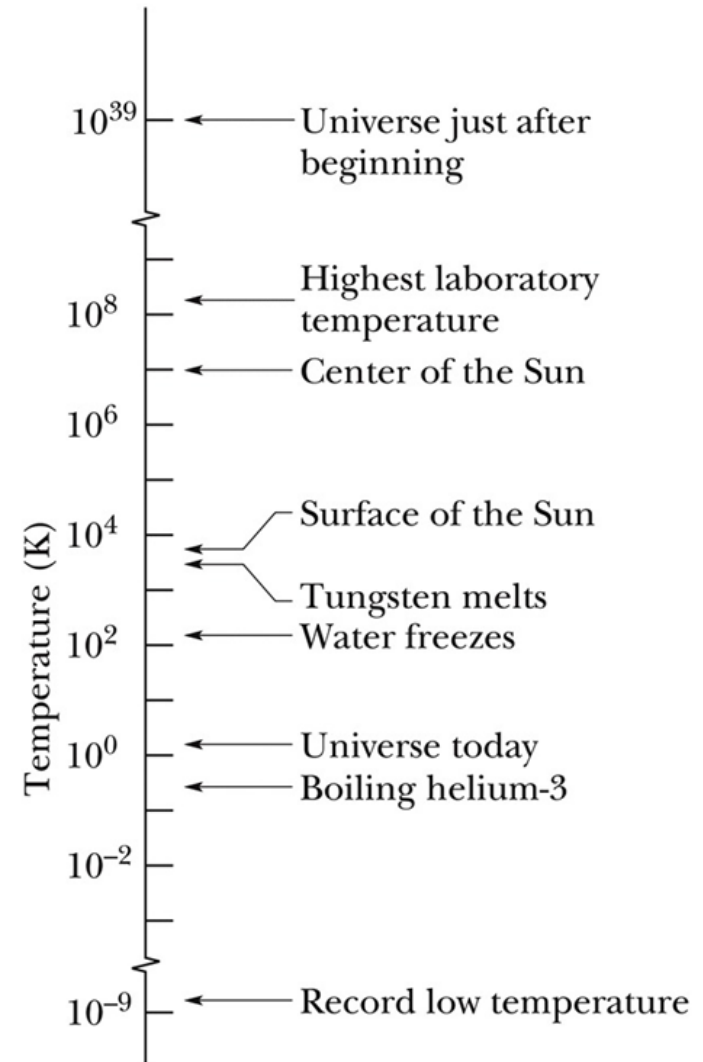


Learning Outcomes

- Explain the temperature system and heat transfer.
- Find the thermal expansion/stress.
- Describe phase transition processes.
- Analyze the heat absorption process.
- Solve problems about the heat transfer.

Temperature

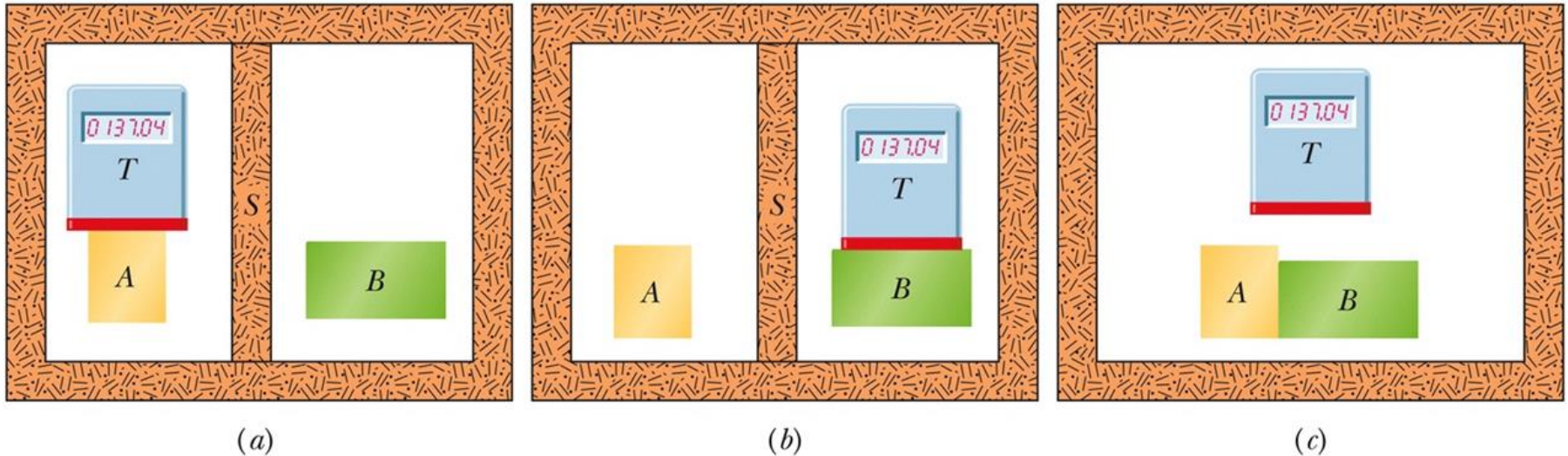
- **Thermodynamics** is the study and application of the thermal energy (often called the internal energy) of systems. One of the central concepts of thermodynamics is temperature.
- **Temperature** is an SI base quantity *related to our sense of hot and cold*. It is measured with a thermometer, which contains a working substance with a measurable property, such as length or pressure, that changes in a regular way as the substance becomes hotter or colder. Physicists measure temperature on the **Kelvin scale**, which is marked in units called kelvins.



Copyright © 2014 John Wiley & Sons, Inc. All rights reserved.

Two bodies are in thermal equilibrium if they are at the same temperature throughout and therefore no heat will flow from one body to the other.

The Zeroth Law of Thermodynamics

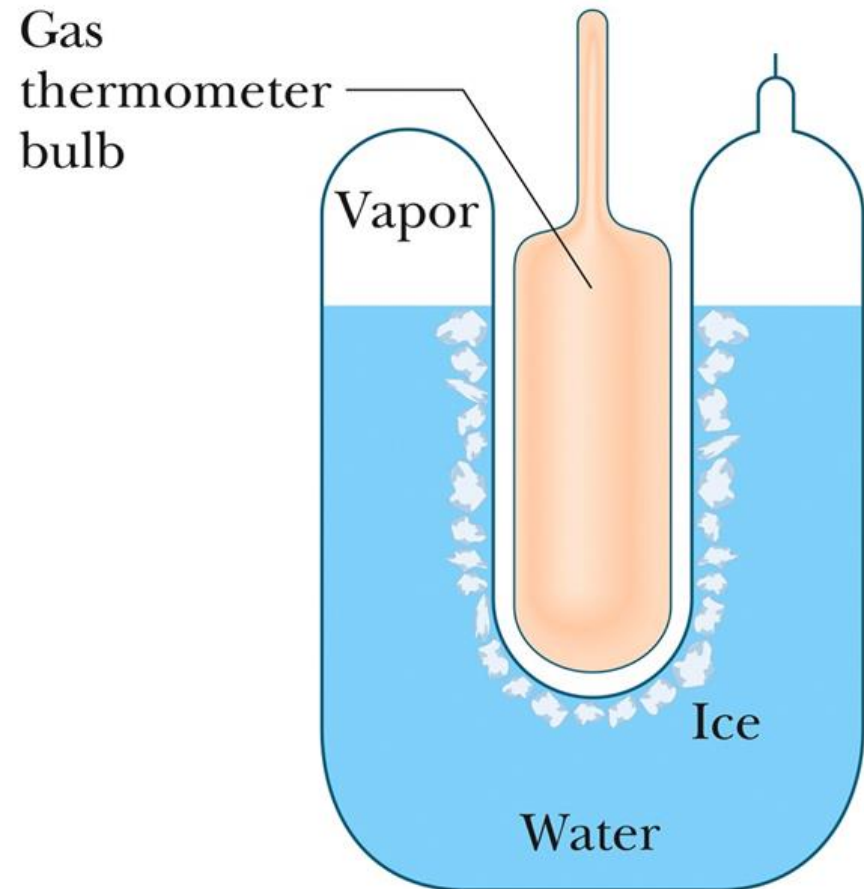


Copyright © 2014 John Wiley & Sons, Inc. All rights reserved.

If bodies A and B are each in thermal equilibrium with a third body T , then A and B are in thermal equilibrium with each other.

Triple Point of Water

- The Triple point of water is the point in which solid ice, liquid water, and water vapor coexist in thermal equilibrium. (*This does not occur at normal atmospheric pressure.*)
- By international agreement, the temperature of this mixture has been defined to be 273.16 K. The bulb of a constant-volume gas thermometer is shown inserted into the well of the cell.



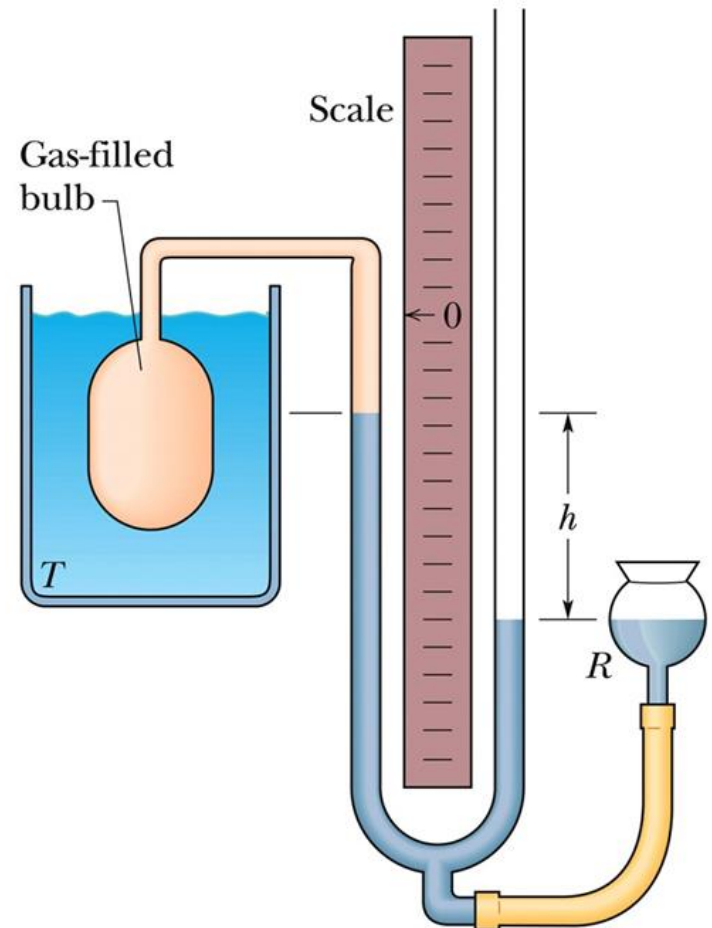
Copyright © 2014 John Wiley & Sons, Inc. All rights reserved.

A triple-point cell

Constant-Volume Gas Thermometer

- It consists of a gas-filled bulb connected by a tube to a mercury manometer. By raising and lowering reservoir R , the mercury level in the left arm of the U-tube can always be brought to the zero of the scale to keep the gas volume constant.
- the recipe for measuring a temperature with a gas thermometer, where p is the observed pressure and p_3 is the pressure at the triple point of water, is

$$T = (273.16 \text{ K}) \left(\lim_{p \rightarrow p_3} \frac{p}{p_3} \right).$$



Copyright © 2014 John Wiley & Sons, Inc. All rights reserved.

Constant-Volume Gas Thermometer

The Celsius and Fahrenheit Scales

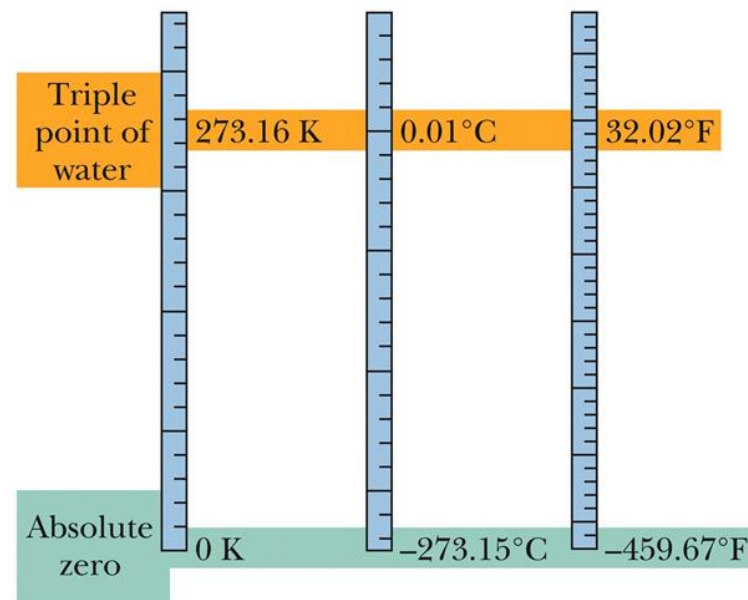
- The **Celsius temperature scale** is defined by

$$T_C = T - 273.15^\circ, \quad \text{with } T \text{ in kelvins.}$$

- The **Fahrenheit temperature scale** is ‘defined’ by

$$T_F = \frac{9}{5} T_C + 32^\circ.$$

Temperature	degree Celsius	Degree Fahrenheit
Boiling point of water ^a	100	212
Normal body temperature	37.0	98.6
Accepted comfort level	20	68
Freezing point of water ^a	0	32
Zero of Fahrenheit scale	≈−18	0
Scales coincide	−40	−40



Copyright © 2014 John Wiley & Sons, Inc. All rights reserved.

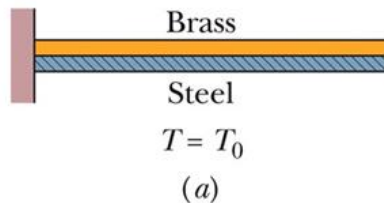
Thermal Expansion

Linear Expansion

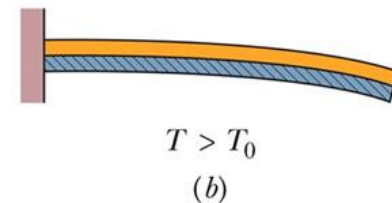
- All objects change size with changes in temperature. For a temperature change ΔT , a change ΔL in any linear dimension L is given by

$$\Delta L = L \alpha \Delta T,$$

in which α is the **coefficient of linear expansion**.

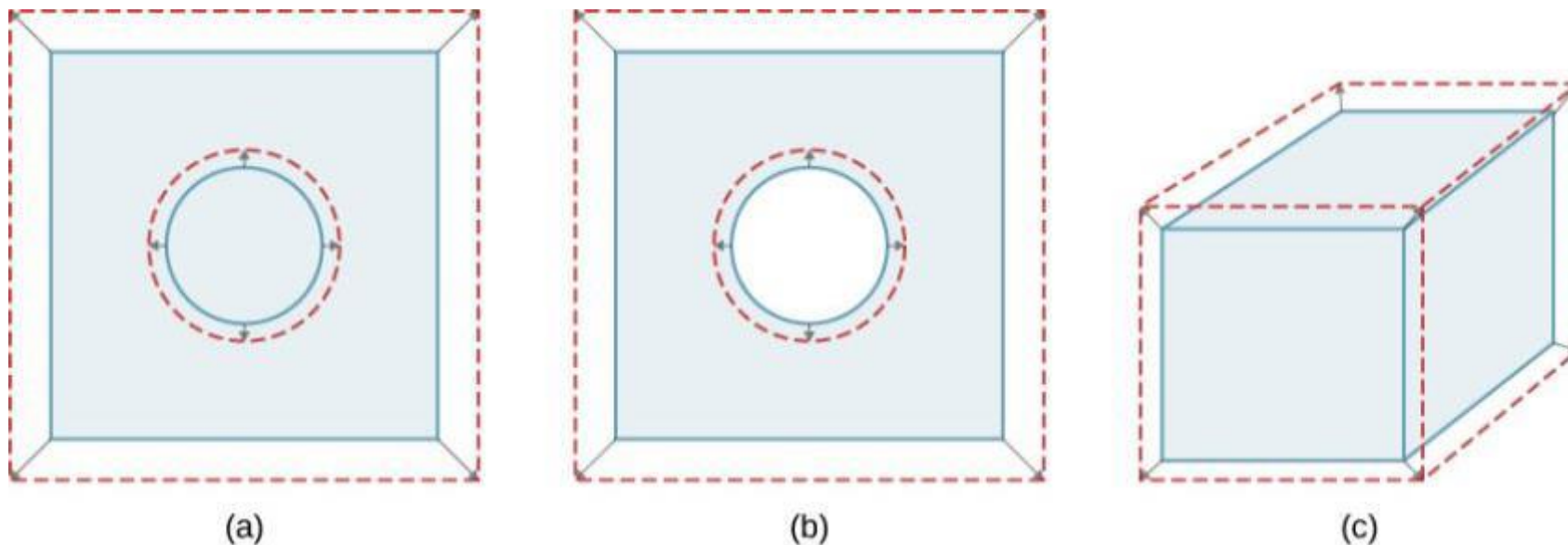


Different amounts of expansion or contraction can produce bending.



Copyright © 2014 John Wiley & Sons, Inc. All rights reserved.

FIGURE 1.7



In general, objects expand in all directions as temperature increases. In these drawings, the original boundaries of the objects are shown with solid lines, and the expanded boundaries with dashed lines.

- (a) Area increases because both length and width increase. The area of a circular plug also increases.
- (b) If the plug is removed, the hole it leaves becomes larger with increasing temperature, just as if the expanding plug were still in place.
- (c) Volume also increases, because all three dimensions increase.

Volume Expansion

- If the temperature of a solid or liquid whose volume is V is increased by an amount ΔT , the increase in volume is found to be $\Delta V = V \beta \Delta T$,

in which β

is the **coefficient of volume expansion** and is related to linear expansion in this way,

$$\beta = 3\alpha.$$

Table 18.3.1 Some Coefficients of Linear Expansion

Substance	α ($10^{-6}/\text{C}^\circ$)
Ice (at 0°C)	51
Lead	29
Aluminum	23
Brass	19
Copper	17
Concrete	12
Steel	11
Glass (ordinary)	9
Glass (Pyrex)	3.2
Diamond	1.2
Invar*	0.7
Fused quartz	0.5

Room temperature values except for the ice listing

*Invar is an alloy that was designed to have a low coefficient of expansion. The word is a shortened form of “invariable.”

Thermal Expansion in Two Dimensions

For small temperature changes, the change in area ΔA is given by

$$\Delta A = 2\alpha A\Delta T \quad (1.3)$$

where ΔA is the change in area A , ΔT is the change in temperature, and α is the coefficient of linear expansion, which varies slightly with temperature. (The derivation of this equation is analogous to that of the more important equation for three dimensions, below.)

Thermal Expansion in Three Dimensions

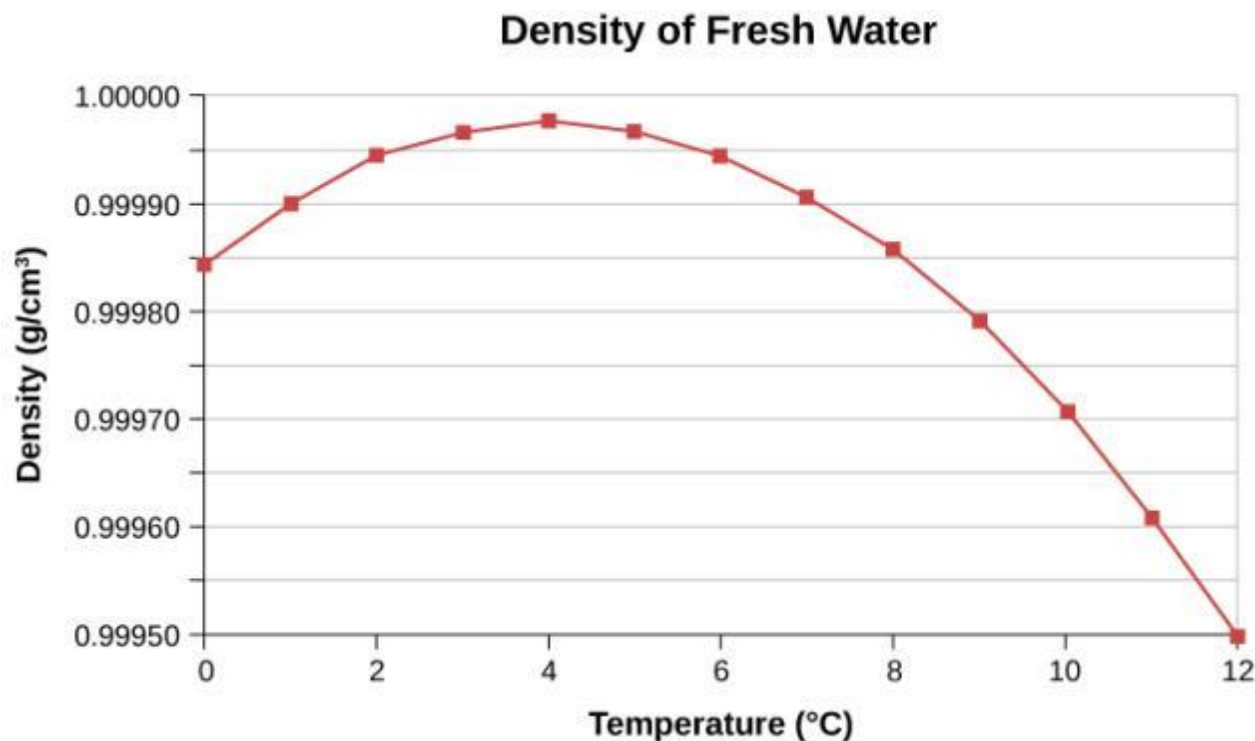
The relationship between volume and temperature $\frac{dV}{dT}$ is given by $\frac{dV}{dT} = \beta V\Delta T$, where β is the **coefficient of volume expansion**. As you can show in **Exercise 1.60**, $\beta = 3\alpha$. This equation is usually written as

$$\Delta V = \beta V\Delta T. \quad (1.4)$$

Note that the values of β in **Table 1.2** are equal to 3α except for rounding.

What about the freezing process of water in a pond?

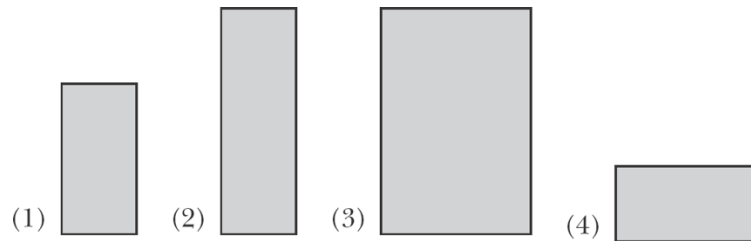
FIGURE 1.8



This curve shows the density of water as a function of temperature. Note that the thermal expansion at low temperatures is very small. The maximum density at 4 °C is only 0.0075% greater than the density at 2 °C, and 0.012% greater than that at 0 °C. The decrease of density below 4 °C occurs because the liquid water approaches the solid crystal form of ice, which contains more empty space than the liquid.

Checkpoint 2

The figure here shows four rectangular metal plates, with sides of L , $2L$, or $3L$. They are all made of the same material, and their temperature is to be increased by the same amount. Rank the plates according to the expected increase in (a) their vertical heights and (b) their areas, greatest first.



Answer:

(a) 2 and 3 (same increase in height), then 1, and then 4

(b) 3, then 2, then 1 and 4 (identical increase in area)

FIGURE 1.5



(a)



(b)

(a) Thermal expansion joints like these in the (b) Auckland Harbour Bridge in New Zealand allow bridges to change length without buckling. (credit: “ŠJů”/Wikimedia Commons)

Example 17-3: Bridge expansion.

The steel bed of a suspension bridge is 200 m long at 20°C . If the extremes of temperature to which it might be exposed are -30°C to $+40^{\circ}\text{C}$, how much will it contract and expand?

Thermal Stresses

A material may be **fixed** at its ends and therefore be unable to expand when the temperature changes. It will then experience **large compressive or tensile stress—thermal stress—**when its temperature changes. The **force required to keep the material from expanding is found from:**

$$\Delta \ell = \frac{1}{E} \frac{F}{A} \ell_0,$$

where E is the Young's modulus of the material. Therefore, the **stress** is:

$$\frac{F}{A} = \alpha E \Delta T.$$

Example: Stress in concrete on a hot day.

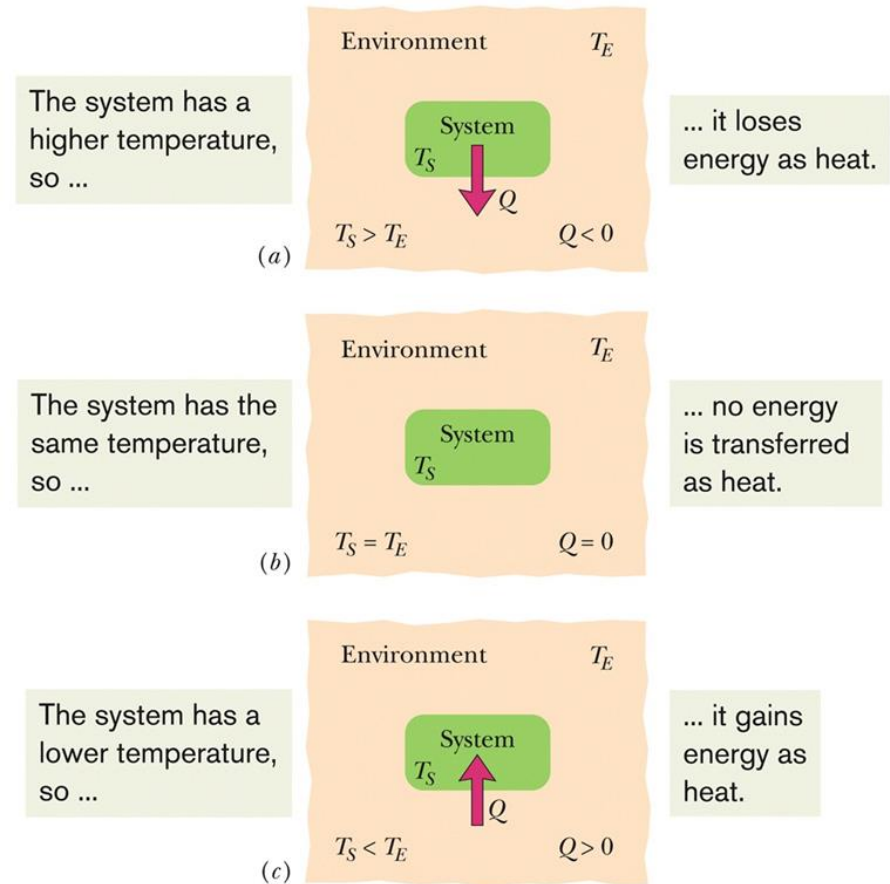
A highway is to be made of blocks of concrete 10 m long placed end to end with no space between them to allow for expansion. If the blocks were placed at a temperature of 10°C , what compressive stress would occur if the temperature reached 40°C ? The contact area between each block is 0.20 m^2 . Will fracture occur?

Absorption of Heat

Temperature and Heat

- Heat Q is energy that is transferred between a system and its environment because of a temperature difference between them.
- It can be measured in joules (J), calories (cal), kilocalories (Cal or kcal), or British thermal units (Btu), with

$$1 \text{ cal} = 3.968 \times 10^{-3} \text{ Btu} = 4.1868 \text{ J}.$$



Copyright © 2014 John Wiley & Sons, Inc. All rights reserved.

Absorption of Heat by Solids and Liquids

- The **heat capacity** C of an object is the proportionality constant between the heat Q that the object absorbs or loses and the resulting temperature change ΔT of the object; that is,

$$Q = C \Delta T = C(T_f - T_i),$$

in which T_i and T_f are the initial and final temperatures of the object. If the object has mass m , then,

$$Q = cm \Delta T = cm(T_f - T_i).$$

where c is the **specific heat** of the material making up the object.

Checkpoint 3

A certain amount of heat Q will warm 1 g of material A by 3 degree Celsius and 1 g of material B by 4 degree Celsius. Which material has the greater specific heat?

Answer:

Material A has the greater specific heat

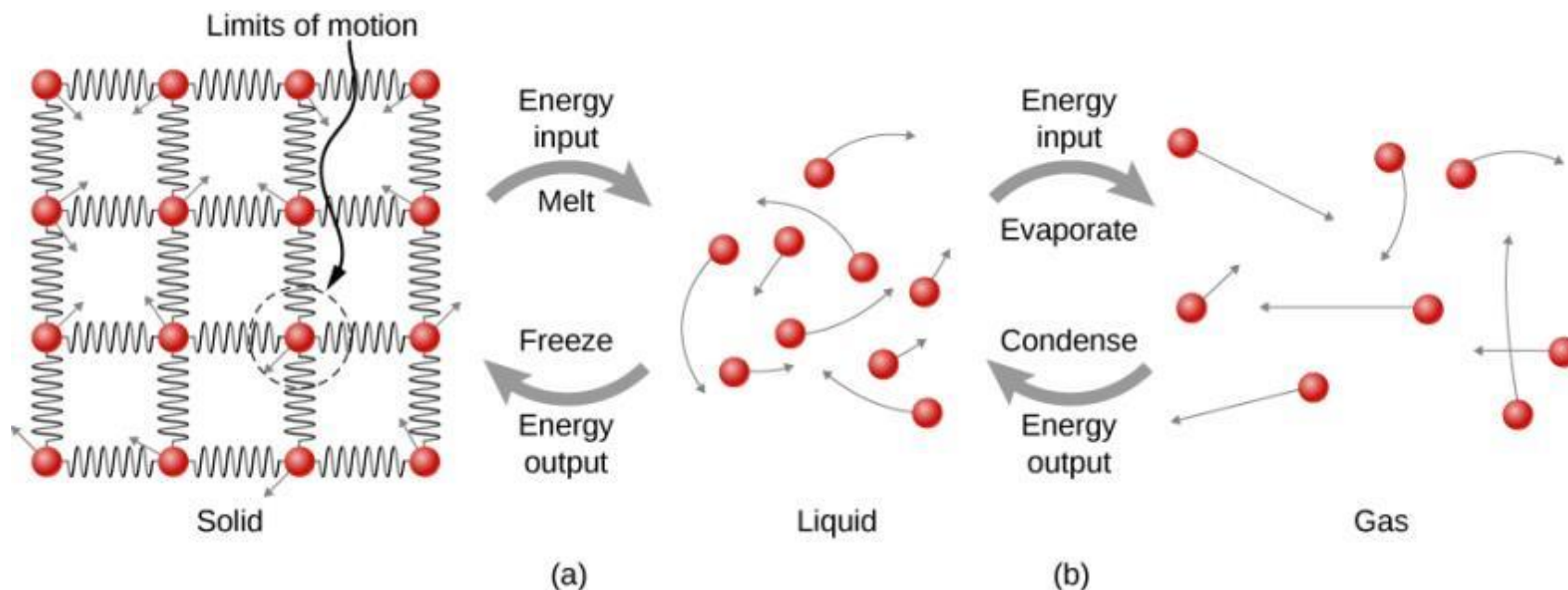
- When quantities are expressed in moles, specific heats must also involve moles (rather than a mass unit); they are then called **molar specific heats**. Table shows the values for some elemental solids (each consisting of a single element) at room temperature.

$$1 \text{ mol} = 6.02 \times 10^{23} \text{ elementary units}$$

- The amount of energy per unit mass that must be transferred as heat when a sample completely undergoes a phase change is called the **heat of transformation (latent heat) L** . Thus, when a sample of mass m completely undergoes a phase change, the total energy transferred is

$$Q = Lm.$$

FIGURE 1.18

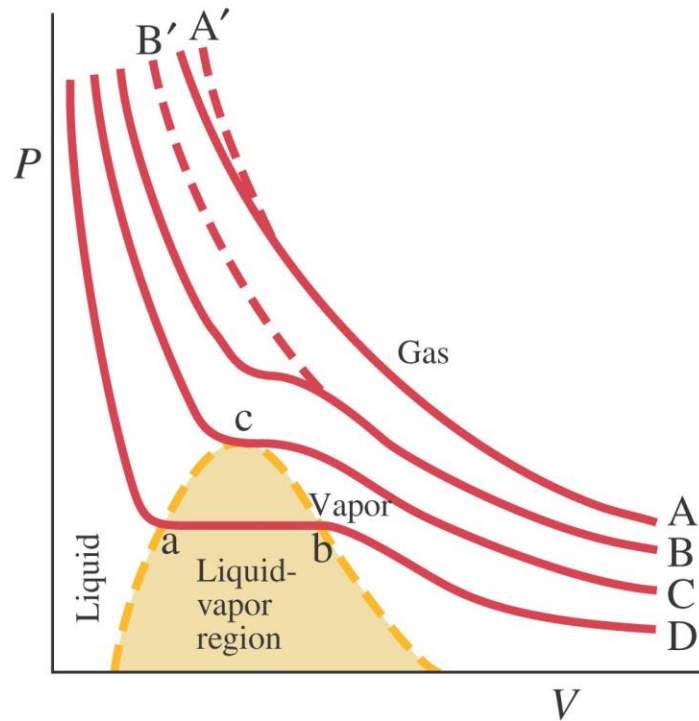


- (a) Energy is required to partially overcome the attractive forces (modeled as springs) between molecules in a solid to form a liquid. That same energy must be removed from the liquid for freezing to take place.
- (b) Molecules become separated by large distances when going from liquid to vapor, requiring significant energy to completely overcome molecular attraction. The same energy must be removed from the vapor for condensation to take place.

Latent heat

Real Gases and Changes of Phase

The curves here represent the behavior of the gas at different temperatures. The cooler it gets, the further the gas is from ideal.



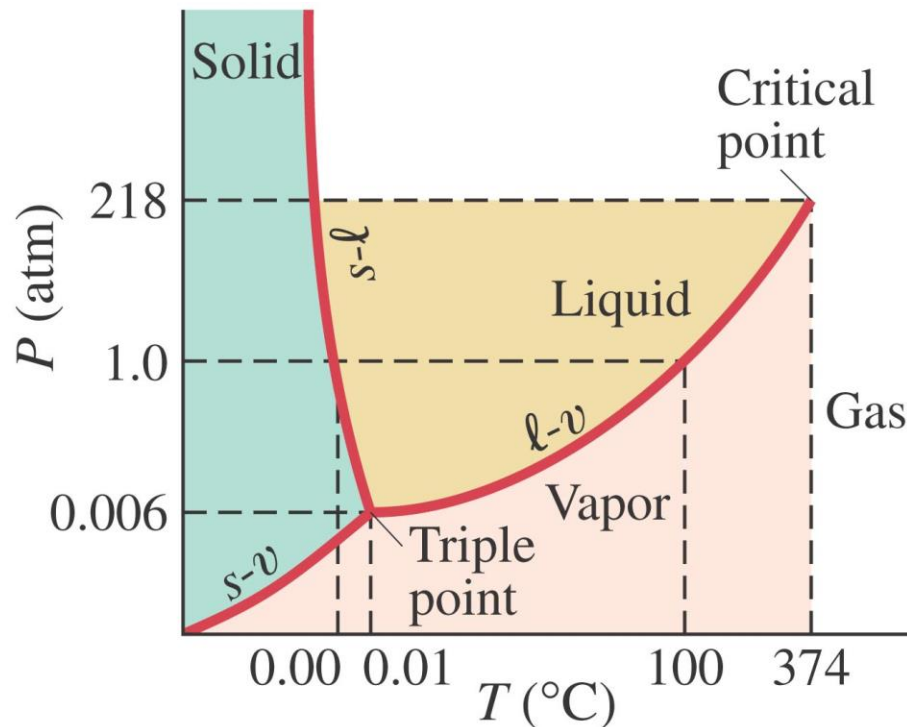
In curve D, the gas becomes **liquid**; it begins condensing at (b) and is entirely liquid at (a). The point (c) is called the **critical point**.

Below the critical temperature, the gas can liquefy if the pressure is sufficient; above it, no amount of pressure will suffice.

TABLE 18–1 Critical Temperatures and Pressures

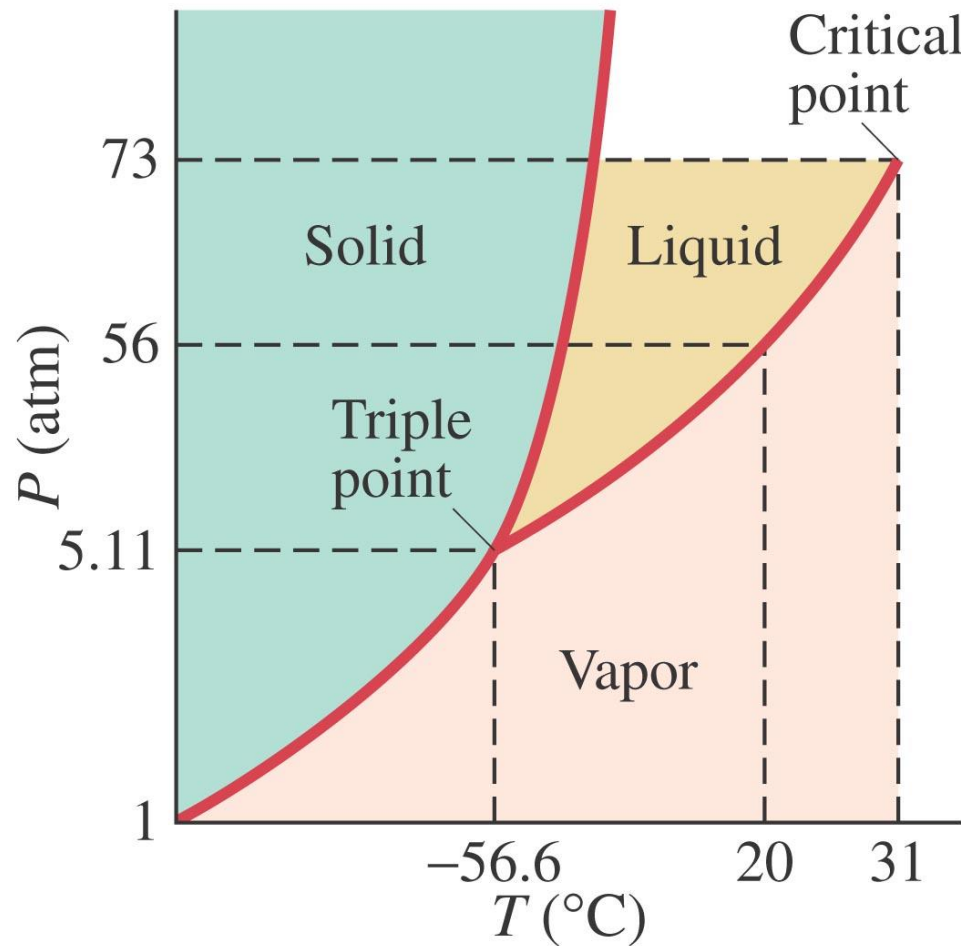
Substance	Critical Temperature		Critical Pressure (atm)
	°C	K	
Water	374	647	218
CO ₂	31	304	72.8
Oxygen	−118	155	50
Nitrogen	−147	126	33.5
Hydrogen	−239.9	33.3	12.8
Helium	−267.9	5.3	2.3

A *PT* diagram is called a **phase diagram**; it shows all three phases of matter. The solid-liquid transition is **melting or freezing**; the liquid-vapor one is **boiling or condensing**; and the solid-vapor one is **sublimation**.



Phase diagram of water.

The triple point is the only point where all three phases can coexist in equilibrium.



Phase diagram of carbon dioxide.

Absorption of Heat

Blank	Specific Heat	Specific Heat	Molar Specific Heat
Blank	Cal	J	J
Substance	g · K	Kg · K	mol · K
Elemental Solids			
Lead	0.0305	128	26.5
Tungsten	0.0321	134	24.8
Silver	0.0564	236	25.5
Copper	0.0923	386	24.5
Aluminum	0.215	900	24.4
Other Solids			
Brass	0.092	380	

Check the textbook for the more substances.

Example: How heat transferred depends on specific heat.

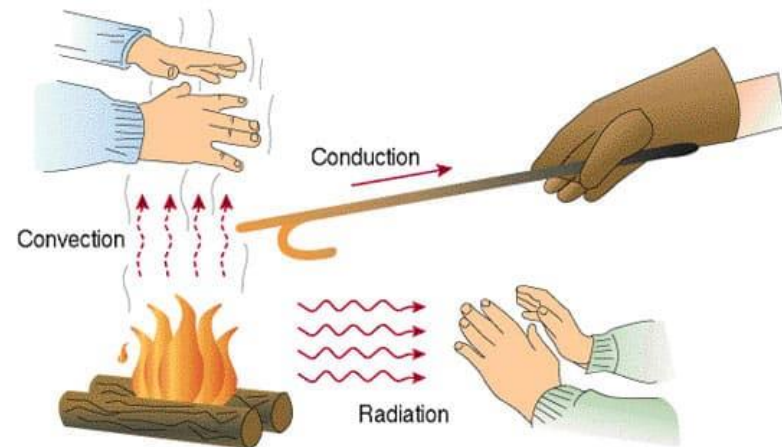
- (a) How much heat input is needed to raise the temperature of an empty 20-kg vat made of iron from 10°C to 90°C ?**
- (b) What if the vat is filled with 20 kg of water?**

Example: Determining a latent heat.

The specific heat of liquid mercury is $140 \text{ J/kg} \cdot ^\circ \text{C}$. When 1.0 kg of solid mercury at its melting point of -39°C is placed in a 0.50-kg aluminum calorimeter filled with 1.2 kg of water at 20.0°C , the mercury melts and the final temperature of the combination is found to be 16.5°C . What is the heat of fusion of mercury in J/kg ?

Convection

- Convection occurs when temperature differences cause an energy transfer by motion within a fluid.
- When you look at the flame of a candle or a match, you are watching thermal energy being transported upward by convection.
- Convection is part of many natural processes. Atmospheric convection plays a fundamental role in determining global climate patterns and daily weather variations. Glider pilots and birds alike seek rising thermals (convection currents of warm air) that keep them aloft. Huge energy transfers take place within the oceans by the same process.



<https://schoolworkhelper.net/thermal-energy-transfer-conduction-convection-radiation/>

Heat Transfer Mechanisms

Thermal Conduction

The rate P_{cond} at which energy is conducted through a slab for which one face is maintained at the higher temperature T_H and the other face is maintained at the lower temperature T_C is

$$P_{\text{cond}} = \frac{Q}{t} = kA \frac{T_H - T_C}{L},$$

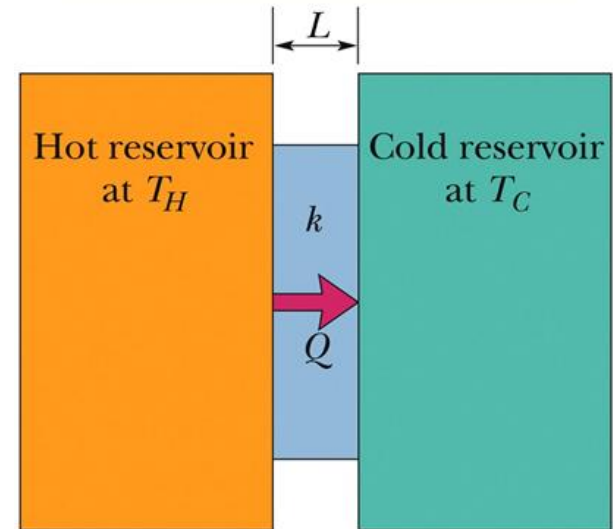
Here each face of the slab has area A , the length of the slab (the distance between the faces) is L , and k is the thermal conductivity of the material.

Energy is transferred as heat from a reservoir at temperature T_H to a cooler reservoir at temperature T_C through a conducting slab of thickness L and thermal conductivity k .



Edward Kinsman/Photo Researchers, Inc.

We assume a steady transfer of energy as heat.



$$T_H > T_C$$

Copyright © 2014 John Wiley & Sons, Inc. All rights reserved.

Thermal Radiation

Radiation is an energy transfer via the emission of electromagnetic energy. The rate P_{rad} at which an object emits energy via thermal radiation is

$$P_{\text{rad}} = \sigma \varepsilon A T^4.$$

Here $\sigma (= 5.6704 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)$ is the Stefan– Boltzmann constant, ε is the emissivity of the object's surface, A is its surface area, and T is its surface temperature (in kelvins). The rate P_{abs} at which an object absorbs energy via thermal radiation from its environment, which is at the uniform temperature T_{env} (in kelvins), is

$$P_{abs} = \sigma \varepsilon A T_{env}^4.$$

$$P_{net} = P_{abs} - P_{rad} = \sigma \varepsilon A (T_{env}^4 - T^4).$$



Learning Outcomes

- ✓ Explain the temperature system and heat transfer. $T_C = T - 273.15^\circ$, $T_F = \frac{9}{5}T_C + 32^\circ$.
- ✓ Find the thermal expansion/stress.
- ✓ Describe phase transition processes.
- ✓ Analyze the heat absorption process.
- ✓ Solve problems about the heat transfer.

$$\Delta L = L \alpha \Delta T, \quad \Delta \ell = \frac{1}{E} \frac{F}{A} \ell_0, \quad \frac{F}{A} = \alpha E \Delta T.$$

$$Q = cm \Delta T = cm(T_f - T_i). \quad P_{\text{cond}} = \frac{Q}{t} = kA \frac{T_H - T_C}{L},$$

CHAPTER REVIEW AND EXAMPLES, DEMOS

<https://openstax.org/books/university-physics-volume-2/pages/1-summary>

... many physical quantities are defined solely in terms of how they are observed or measured, that is, they are defined *operationally*. **Temperature** is operationally defined as the quantity of what we measure with a thermometer.

Any physical property that depends consistently and reproducibly on temperature can be used as the basis of a thermometer.

... in thermodynamics, we require a system to be macroscopic, that is, to consist of a huge number (such as 10^{23}) of molecules. Then we can say that a system is in thermal equilibrium with itself if all parts of it are at the same temperature.

The definition of temperature in terms of molecular motion suggests that there should be a lowest possible temperature, where the average kinetic energy of molecules is zero (or the minimum allowed by quantum mechanics). Experiments

If you cut a hole in a metal plate, the remaining material will expand exactly as it would if the piece you removed were still in place. The piece would get bigger, so the hole must get bigger too.

Example 1.3

Calculating Thermal Expansion

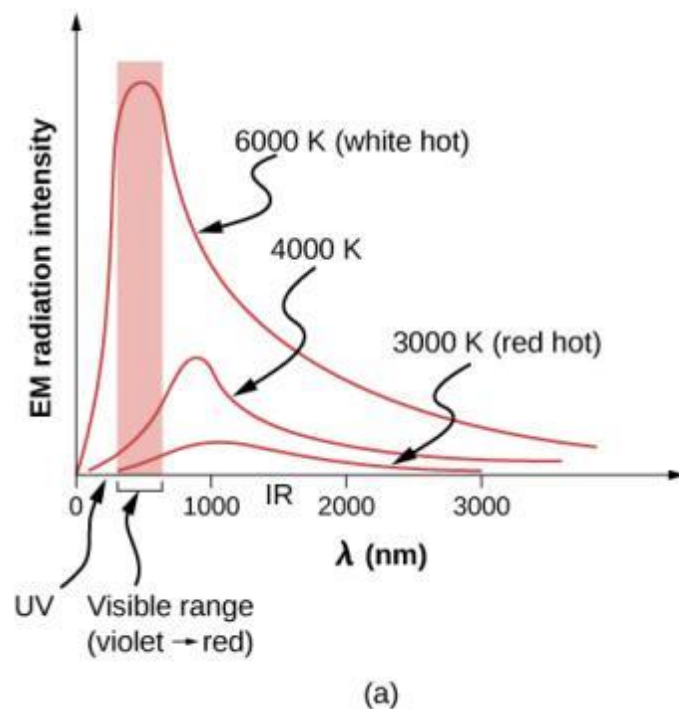
Suppose your 60.0-L (15.9 -gal -gal) steel gasoline tank is full of gas that is cool because it has just been pumped from an underground reservoir. Now, both the tank and the gasoline have a temperature of 15.0 °C. How much gasoline has spilled by the time they warm to 35.0 °C ?

Example 1.8

Temperature-Dependent Heat Capacity

At low temperatures, the specific heats of solids are typically proportional to T^3 . The first understanding of this behavior was due to the Dutch physicist Peter Debye, who in 1912, treated atomic oscillations with the quantum theory that Max Planck had recently used for radiation. For instance, a good approximation for the specific heat of salt, NaCl, is $c = 3.33 \times 10^4 \frac{\text{J}}{\text{kg} \cdot \text{K}} \left(\frac{T}{321 \text{ K}} \right)^3$. The constant 321 K is called the *Debye temperature* of NaCl, Θ_D , and the formula works well when $T < 0.04\Theta_D$. Using this formula, how much heat is required to raise the temperature of 24.0 g of NaCl from 5 K to 15 K?

FIGURE 1.29



- (a) A graph of the spectrum of electromagnetic waves emitted from an ideal radiator at three different temperatures. The intensity or rate of radiation emission increases dramatically with temperature, and the spectrum shifts down in wavelength toward the visible and ultraviolet parts of the spectrum. The shaded portion denotes the visible part of the spectrum. It is apparent that the shift toward the ultraviolet with temperature makes the visible appearance shift from red to white to blue as temperature increases.
- (b) Note the variations in color corresponding to variations in flame temperature.

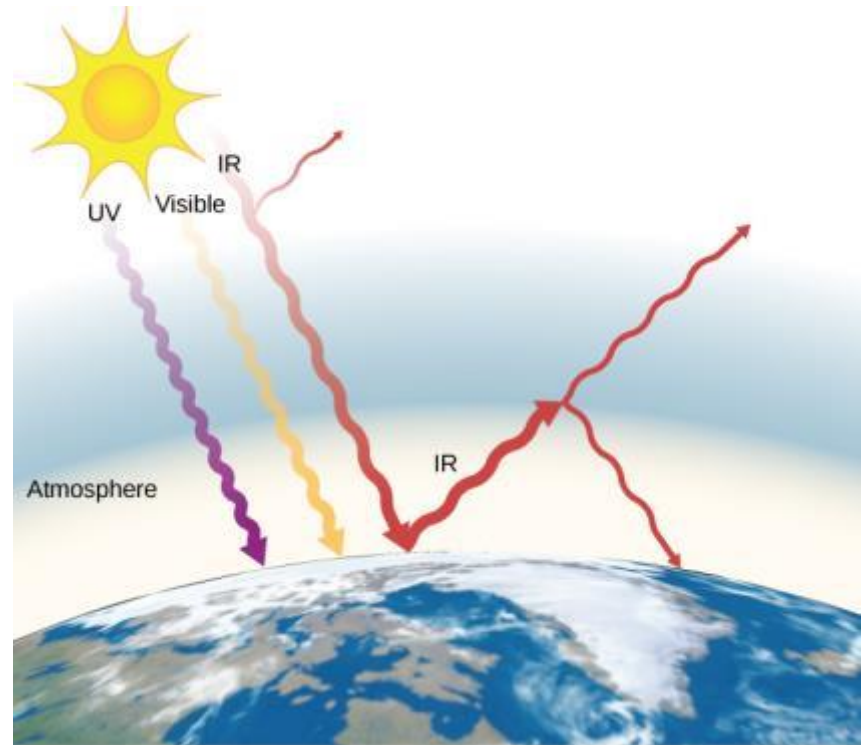
FIGURE 1.32



A thermograph of part of a building shows temperature variations, indicating where heat transfer to the outside is most severe. Windows are a major region of heat transfer to the outside of homes.
(credit: US Army)

FIGURE 1.33

The greenhouse effect is the name given to the increase of Earth's temperature due to absorption of radiation in the atmosphere. The atmosphere is transparent to incoming visible radiation and most of the Sun's infrared. The Earth absorbs that energy and re-emits it. Since Earth's temperature is much lower than the Sun's, it re-emits the energy at much longer wavelengths, in the infrared. The atmosphere absorbs much of that infrared radiation and radiates about half of the energy back down, keeping Earth warmer than it would otherwise be. The amount of trapping depends on concentrations of trace gases such as carbon dioxide, and an increase in the concentration of these gases increases Earth's surface temperature.



Questions

